SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2023	**Date:**	

Code of Conduct

I B.Jahnavi / 192325043 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: b.jahnavi

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences
Failure to handle subject–verb agreement
Inefficient parsing for long queries
Example user query:
"Show me the transactions with the card from last month"
a) Analyze the ambiguity present in the given sentence using CFG
b) Evaluate the limitations of CFG in handling real-world conversational queries
c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:
Backtracking issues
Poor handling of incomplete or ambiguous input
Difficulty in real-time response
Example command:
"Book a flight to Delhi with a window seat"
a) Analyze how the given sentence can lead to multiple parse structures
b) Analyze the limitations of top-down parsing in this real-time application
c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs
d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.
The system must handle:
Complex and lengthy medical sentences
Ambiguity in sentence structure
Accurate subject-verb agreement
Relationships between medical actions and entities (sub-categorization)
Real-time processing for hospital use
Example input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.
a) Design a complete NLP architecture integrating:
CFG for syntactic structure
Probabilistic CFG (PCFG) for ambiguity resolution
Feature Structures for agreement handling
Efficient parsing techniques
b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).
c) Create a method to:
Incorporate sub-categorization frames for medical verbs
Handle real-time data processing
Improve accuracy and scalability in a hospital environment

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Q1. Banking Chatbot — "Show me the transactions with the card from last month"

a) Ambiguity Present in the Sentence Using CFG

This is a classic prepositional-phrase (PP) attachment ambiguity. CFG rewrite rules allow at least two valid parse trees for the same sentence:

- Show me [the transactions [with the card] [from last month]] — both PPs modify "transactions" (transactions made using the card, during last month).
- Show me [the transactions [with the card [from last month]]] — "from last month" modifies "the card" (implying a card issued last month).

CFG rules only enforce syntactic well-formedness, so both trees are equally "grammatical" and the parser has no built-in way to decide which structure reflects the customer's actual intent.

b) Limitations of CFG in Real-World Conversational Queries

- No semantic plausibility scoring — CFG cannot rank competing attachments.
- No feature/agreement checking — subject-verb agreement is not encoded in plain CFG rules.
- Exponential growth of parse trees as prepositional/clause chains lengthen, hurting parsing efficiency.
- Treats all grammatically valid parses as equally likely, with no statistical preference.
- Brittle with fragments, disfluencies, or unseen banking-domain terminology.

c) Proposed Improved Solution — PCFG, Feature Structures, and Earley Parsing

- PCFG: assigns probabilities to grammar rules learned from real query logs, so the statistically common attachment (both PPs modifying "transactions") is automatically preferred.
- Feature Structures: attach attribute-value pairs (number, tense, agreement) to parse nodes and use unification to reject grammatically inconsistent parses.
- Earley Parsing: a chart-based, dynamic-programming parser that handles any CFG — including ambiguous or left-recursive grammars — in polynomial time by sharing sub-computations instead of re-deriving them.

d) Justification of Improved Accuracy and Efficiency

- Efficiency: Earley parsing avoids the combinatorial blow-up of naive parsing, enabling real-time responses even for long, complex queries.

- Accuracy: PCFG resolves ambiguity by selecting the most probable interpretation instead of presenting all parses as equally valid.
- Robustness: feature structures catch grammatical inconsistencies arising from noisy, conversational input.
- Together, this hybrid pipeline scales to production query volumes while reducing misrouted banking requests caused by misinterpreted intent.

Q2. Voice Assistant — "Book a flight to Delhi with a window seat"

a) Multiple Parse Structures

The phrase "with a window seat" can attach at different points in the parse tree, producing multiple structurally valid readings:

- Attached to "flight" → "a flight that has a window seat" (the intended meaning).
- Attached to "book" (the verb) → the booking action is performed "using a window seat," which is syntactically legal but semantically nonsensical.
- Attached to "Delhi" → an unlikely but grammatically possible reading where the PP modifies the destination.

All three are valid under CFG rules, but only the first is semantically sensible — illustrating that syntax alone cannot resolve real-world attachment ambiguity.

b) Limitations of Top-Down Parsing in Real-Time Applications

- Backtracking: the parser retries alternative rule expansions on failure, which is costly for real-time voice response.
- Left recursion: naive top-down parsers loop infinitely on left-recursive grammar rules.
- Poor handling of incomplete input: top-down parsing expects a full derivation from the start symbol, so a pause mid-utterance causes failure rather than yielding a usable partial parse.
- Ambiguity handling: exponential backtracking cost when multiple rule choices are simultaneously viable.

c) How Earley Parsing Improves Handling of Ambiguity and Partial Input

- Processes input incrementally, left to right, maintaining the set of valid partial parse states after each word — ideal for streaming speech recognition.
- Uses a dynamic-programming chart to avoid recomputation, giving polynomial-time worst-case complexity (often near-linear in practice).
- Natively handles left-recursive and ambiguous grammars without requiring grammar rewriting.
- Produces a compact, shared parse forest containing all valid interpretations, which can be disambiguated later using probabilistic or semantic scoring.

d) Top-Down vs Earley Parsing — Performance Comparison

Aspect	Top-Down Parsing	Earley Parsing
Time Complexity	Exponential (worst case)	Polynomial $O(n^3)$, often near-linear
Left Recursion	Fails / needs grammar rewriting	Handled natively

Ambiguity	One parse at a time, heavy backtracking	Shared forest of all parses
Real-time / Partial Input	Poor	Well-suited (incremental)

Verdict: Earley parsing is the better fit for real-time, ambiguous, partially spoken voice commands, since it avoids exponential backtracking and natively supports incremental, partial-input processing.

Q3. Healthcare NLP Architecture for Medical Report Analysis

a) Integrated NLP Architecture

- Tokenization and Part-of-Speech tagging of the input medical sentence.
- CFG: builds candidate syntactic parses, capturing embedded structures such as relative clauses (e.g., "who reviewed the patient last week").
- PCFG: ranks the candidate parses by learned probability to resolve structural ambiguity (e.g., deciding where "last week" attaches).
- Feature Structures: unification enforces subject-verb agreement and overall grammatical consistency across the parse.
- Sub-categorization-based semantic role extraction: maps verbs such as "recommend" and "schedule" to their required arguments.
- Efficient parser (Earley/chart-based): polynomial-time, incremental parsing suited to hospital-scale throughput.
- Structured output layer: emits diagnosis / action / location frames for consumption by the Electronic Health Record (EHR) system.

b) Step-by-Step Workflow for the Example Sentence

Sentence: "The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

- Step 1 — Tokenize and POS-tag the sentence.
- Step 2 — CFG parse: identify the main clause "The doctor ... recommends X and Y" with an embedded relative clause "who reviewed the patient last week" modifying "doctor."
- Step 3 — PCFG disambiguation: resolve that "last week" attaches to "reviewed" inside the relative clause (the higher-probability reading), not to "recommends."
- Step 4 — Feature unification: confirm "doctor" (third-person singular) agrees with "recommends" — agreement check passes.
- Step 5 — Sub-categorization: identify that "recommends" takes two coordinated gerund-phrase complements — "starting medication" and "scheduling a follow-up visit in Chennai."
- Step 6 — Role extraction: Agent = Doctor; Recommended Actions = [Start medication, Schedule follow-up visit]; Location = Chennai.
- Step 7 — Structured output: {Agent: "Doctor", RecommendedActions: ["Start medication", "Schedule follow-up visit"], Location: "Chennai"}.

c) Sub-Categorization, Real-Time Processing, and Scalability

- Build a domain-specific verb lexicon (prescribe, recommend, schedule, diagnose), each tagged with its expected argument frame, to constrain valid semantic-role assignments.

- Use incremental/streaming Earley parsing with caching of frequently occurring clinical sentence templates to reduce latency.
- Continuously retrain PCFG probabilities and sub-categorization frames on growing annotated hospital corpora, using active learning from corrected outputs.
- Scale the pipeline as horizontally-deployed microservices to support concurrent, department-wide real-time processing across the hospital.